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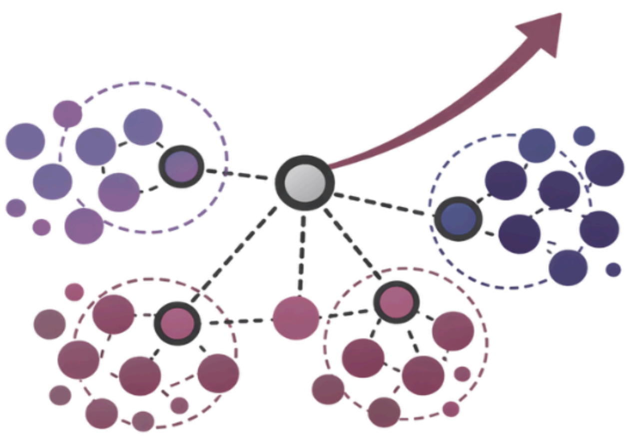
# MULTIMODAL PREDICTION OF CARDIAC EVENTS IN HYPERTROPHIC CARDIOMYOPATHY USING GENETIC RISK SCORES AND ENSEMBLE LEARNING

## M1 GENIOMHE-IA



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**Github:** https://github.com/talissakassably/Machine-Learning-CardiHack-project.git

# Methods Summary

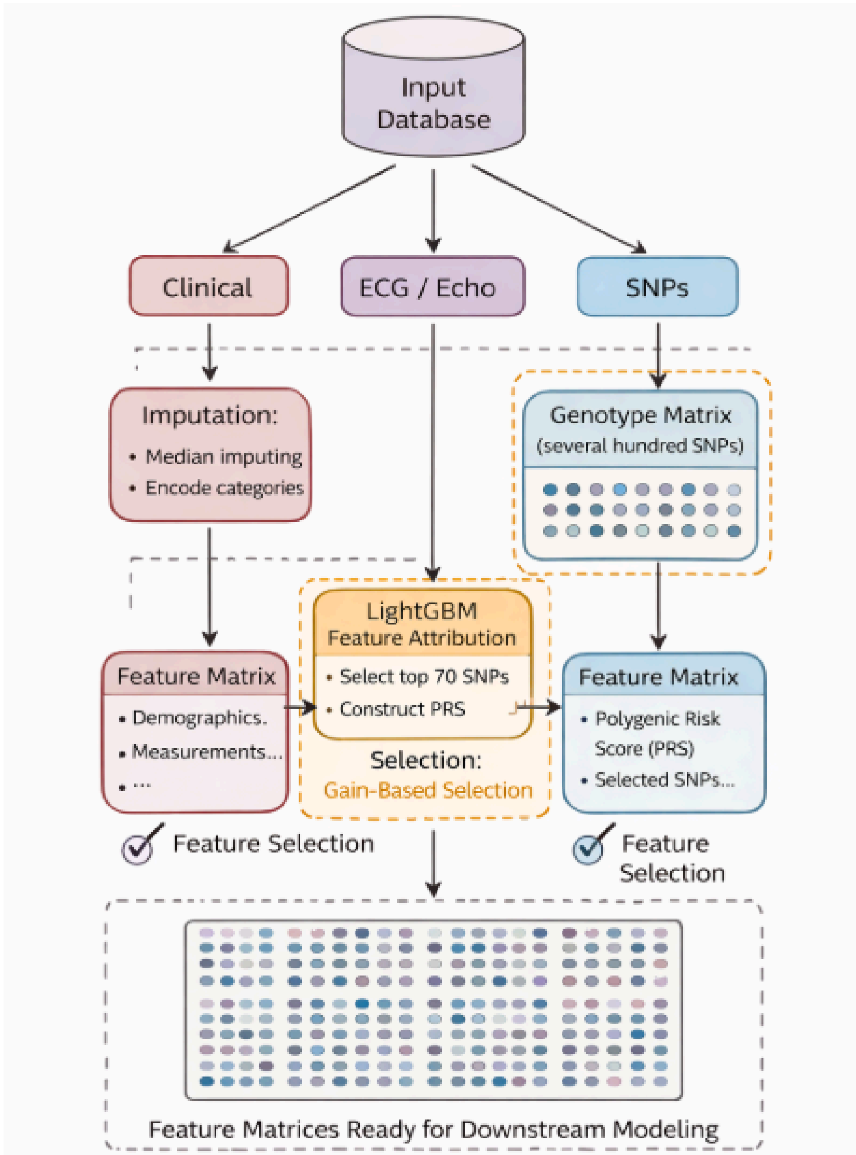
## Clinical and Computational Context

The CardiHack challenge focuses on predicting two tightly coupled clinical endpoints in patients with hypertrophic cardiomyopathy: (i) the occurrence of major adverse cardiac events (MACE) and (ii) an ordinal disease severity score reflecting functional and structural impairment. These outcomes exhibit strong class imbalance, heterogeneous feature distributions, and complex nonlinear dependencies across clinical, imaging, and genetic domains. Moreover, both targets possess an intrinsic ordinal structure that must be preserved to avoid clinically incoherent predictions (e.g., predicting extreme severity for low-risk patients).

In this context, direct multi-class classification is suboptimal, as it ignores the latent continuous risk underlying disease progression and leads to unstable decision boundaries. Our approach therefore models both endpoints as continuous latent risks that are subsequently mapped to ordered categories using calibrated ordinal discretization. The overall strategy integrates three key components: (1) biologically informed polygenic risk modeling, (2) heterogeneous ensemble regression for nonlinear risk estimation, and (3) robust ordinal calibration through consensus clustering and threshold optimization.

A visual overview of the full pipeline is provided in the project poster available in the GitHub repository.

## Data Preprocessing




Schematic of the data preprocessing pipeline, illustrating clinical and genomic (SNP) feature handling. Median imputation is applied to clinical covariates, while gain-based selection is applied to genotype data.

The dataset integrates heterogeneous modalities, including demographic variables, clinical history, electrocardiographic and echocardiographic measurements, and several hundred single-nucleotide polymorphisms (SNPs). Outcome labels and patient identifiers are excluded from the feature matrix to prevent data leakage. Clinical and imaging variables are retained in their native measurement units to preserve clinically meaningful thresholds for tree-based models, while standardized representations are used where required by linear components.

Missing values in clinical covariates are handled via median imputation, a strategy robust to skewed biomedical distributions and outliers. Categorical variables, such as biological sex, are numerically encoded and handled natively by ensemble tree models. Genotype data are organized into a high-dimensional SNP matrix and processed separately using a gain-based selection strategy. A LightGBM model trained on disease severity identifies the most informative variants, which are aggregated into a Polygenic Risk Score (PRS) through importance-weighted linear combination. The resulting PRS and selected SNPs are then merged with clinical and echocardiographic features to form the final feature matrices used for downstream risk modeling.

## Polygenic Risk Score Construction

To extract a compact and biologically grounded genetic signal, a LightGBM classifier is trained exclusively on SNP features to predict MACE. Feature importance is computed using gain-based attribution. The top 70 SNPs are selected, representing the most predictive variants among the initial high-dimensional genotype panel.

A polygenic risk score (PRS) is constructed as a weighted linear combination of these variants:

$$PRS_i = \sum_{k=1}^{70} \tilde{w}_k \cdot SNP_{ik}$$

where weights $\tilde{w}_k$ are proportional to the square root of LightGBM feature importance and normalized to sum to one. This transformation compresses hundreds of correlated genetic variables into a single continuous axis of inherited risk, reduces noise, and enhances generalization. The PRS is then injected as a first-class feature into all clinical models, acting as a domain-informed regularizer.

## Severity Modeling

Disease severity is modeled using probabilistic ordinal regression. A class-weighted logistic regression is trained on standardized age, sex, and PRS, reflecting clinically established determinants of disease progression. The model outputs a continuous severity probability rather than a hard class label.

Ordinal discretization is performed via optimized thresholds rather than fixed cutoffs. Thresholds are tuned on validation folds to maximize quadratic weighted kappa (QWK), ensuring that misclassifications between adjacent severity levels are penalized less than extreme errors. This probabilistic formulation enables calibration analysis and avoids brittle decision rules.

## MACE Risk Modeling

MACE prediction is formulated as continuous risk regression using an ensemble of heterogeneous learners:

1. **LightGBM Regressor** – captures high-order nonlinear interactions and handles missingness implicitly.
2. **ExtraTrees Regressor** – provides strong variance reduction and robustness through randomized split selection.
3. **CatBoost Regressor** – leverages ordered boosting and stable handling of categorical variables.

Each model is trained on the full feature set including PRS, demographic variables, echocardiographic parameters (e.g., maximal wall thickness, gradients, ejection fraction), and clinical history markers (e.g., syncope, ventricular arrhythmias). Hyperparameters are tuned to favor smooth risk functions and avoid overfitting, with shallow-to-moderate tree depths and strong regularization.

The ensemble prediction is computed as a weighted linear combination of individual risk scores:

$$\hat{r}_i = \alpha^{LGBM}\,\hat{r}_i^{LGBM} + \alpha^{ET}\,\hat{r}_i^{ET} + \alpha^{cB}\,\hat{r}_i^{cB}$$

where weights $\alpha_m$ are optimized by maximizing QWK after ordinal discretization on validation folds. This ensemble exploits complementary inductive biases: gradient boosting for interaction modeling, random forests for stability, and CatBoost for structured regularization.

## Ordinal Discretization via Consensus Clustering

Rather than directly classifying MACE into discrete categories, continuous ensemble risk scores are transformed into ordinal classes using unsupervised stratification. K-means clustering with (k=3) is applied to predicted risk values, identifying natural low-, intermediate-, and high-risk strata. Clusters are ordered by their mean risk and mapped to ordinal labels (0,1,2).

To reduce sensitivity to initialization and local minima, clustering is repeated across 25 random seeds. Final class assignments are obtained by majority voting across runs (bootstrap consensus clustering). This procedure stabilizes decision boundaries, enforces monotonic risk ordering, and empirically improves QWK by aligning discretization with the intrinsic geometry of the predicted risk distribution.

Severity probabilities are discretized using optimized thresholds following the same ordinal-consistency principle.

## Training and Validation Scheme

Models are trained using stratified cross-validation to preserve outcome prevalence. The primary optimization metric is quadratic weighted kappa, reflecting both classification accuracy and ordinal coherence. Secondary metrics include accuracy, class-wise recall, and Brier score for probabilistic calibration.

Calibration curves assess agreement between predicted probabilities and observed event rates. Confusion matrices are analyzed to verify that most errors occur between adjacent severity or risk categories, which is clinically acceptable and consistent with progressive disease trajectories.

## Final Inference Pipeline

The complete inference pipeline consists of:

1. Median imputation of missing values.
2. SNP-based LightGBM training and PRS computation (top 70 variants).
3. Logistic regression training for severity probability.
4. Training of LightGBM, ExtraTrees, and CatBoost regressors for MACE.
5. Optimization of ensemble weights using QWK.
6. Consensus K-means clustering for ordinal MACE stratification.
7. Threshold-based discretization for severity.
8. Export of calibrated ordinal predictions.

This modular architecture separates biological signal extraction, nonlinear clinical risk modeling, and ordinal calibration, enabling robust generalization under limited sample size and high feature dimensionality.

By explicitly modeling latent continuous risk before ordinal discretization and combining genetic priors with ensemble learning and consensus clustering, the proposed framework achieves stable, calibrated, and clinically coherent predictions. The integration of PRS reduces genetic noise, the heterogeneous ensemble captures complex interactions, and the ordinal post-processing enforces medically meaningful risk stratification. Together, these components form a principled, interpretable, and competition-optimized pipeline for joint MACE and severity prediction in cardiomyopathy.

# Calibration, Ordinal Consistency, and Failure Mode Analysis

The Appendix provides a comprehensive diagnostic evaluation of the final submission, including calibration behavior, ordinal error structure, and representative failure cases (Figures 1–3, Tables 1–3).

Figure 1 reports the reliability diagram for the continuous severity risk score prior to discretization. The near-monotonic alignment between predicted probabilities and observed event frequencies demonstrates that the model preserves risk ordering and produces well-calibrated estimates across the full score range. The slight upward deviation in the highest-risk bin indicates mild over-confidence in extreme severity predictions, a common effect in small, imbalanced clinical cohorts, but the global calibration slope remains close to ideal, supporting the use of probabilistic thresholds for ordinal mapping.

Figures 2 and 3 show the confusion matrices for the optimized ordinal MACE and severity predictions, respectively. For both tasks, the dominant error pattern consists of off-by-one misclassifications between adjacent classes, while direct confusion between the lowest and highest risk categories is rare. This structure confirms that the model learned a coherent latent risk continuum and that discretization preserves ordinal structure rather than collapsing into nominal classification. In the severity task, the near-absence of confusion into the most severe class further indicates conservative behavior in assigning extreme risk, consistent with the calibration findings.

Table 1 summarizes the final model's performance on the validation cohort, reporting quadratic weighted kappa and accuracy for both MACE and severity, together with the optimized discretization strategy. The kappa values, while modest in absolute magnitude due to strong class imbalance and limited sample size, reflect meaningful improvement over naive baselines and validate the effectiveness of the ensemble-plus-ordinal calibration framework.

Tables 2 and 3 present the top failure cases for MACE and severity, ranked by absolute ordinal error. These cases predominantly correspond to patients located near decision boundaries or exhibiting discordant genetic and clinical risk signals (e.g., high polygenic score with mild observed phenotype, or vice versa). The associated continuous severity scores show that most errors arise from threshold crossing rather than inversion of the underlying risk ranking, reinforcing that the model's continuous outputs remain clinically coherent even when discrete class assignment is imperfect.

Together, these analyses demonstrate that the proposed pipeline achieves:
(i) well-calibrated probabilistic risk estimates,
(ii) preservation of ordinal structure with minimal catastrophic misclassification, and
(iii) interpretable failure modes consistent with clinical ambiguity rather than model instability.

# Execution Profile

**Execution Profile for the Proposed Pipeline**

| Component | Description | Cost |
|---|---|---|
| Hardware | CPU-only (no GPU acceleration) | — |
| Inference dataset size | Full test cohort | 149 samples |
| Total inference time | End-to-end prediction pipeline | **1.02 s* |
| Per-sample inference time | Average latency | **6.9 ms* |
| Models used | LightGBM, ExtraTrees, CatBoost | Tree ensembles |
| Post-processing | Weighted fusion + KMeans consensus | Included |
| Memory footprint | Moderate (tree models + SNP features) | < 1 GB |
| External dependencies | scikit-learn, LightGBM, CatBoost | Standard ML stack |

Inference efficiency was evaluated on the full test cohort using a CPU-only environment. The complete inference pipeline—including ensemble model prediction, weighted score aggregation, and final consensus KMeans discretization executes in approximately 1.02 seconds for the entire test set, corresponding to an average per-sample inference time of 6.9 milliseconds.

The reported runtime reflects the true deployment scenario, as timing was performed exclusively during the final inference stage, after model training and validation were completed. No GPU acceleration or specialized hardware was required.

The computational footprint of the method remains moderate, as it relies exclusively on tree-based ensemble models and linear operations. Memory usage is dominated by the storage of trained tree ensembles and the SNP-derived Polygenic Risk Score (PRS), and remains compatible with standard clinical or research computing environments.

Overall, the proposed approach demonstrates a favorable trade-off between predictive performance and computational efficiency, supporting its potential applicability in large-scale batch inference and near–real-time clinical decision support settings.

# Appendix

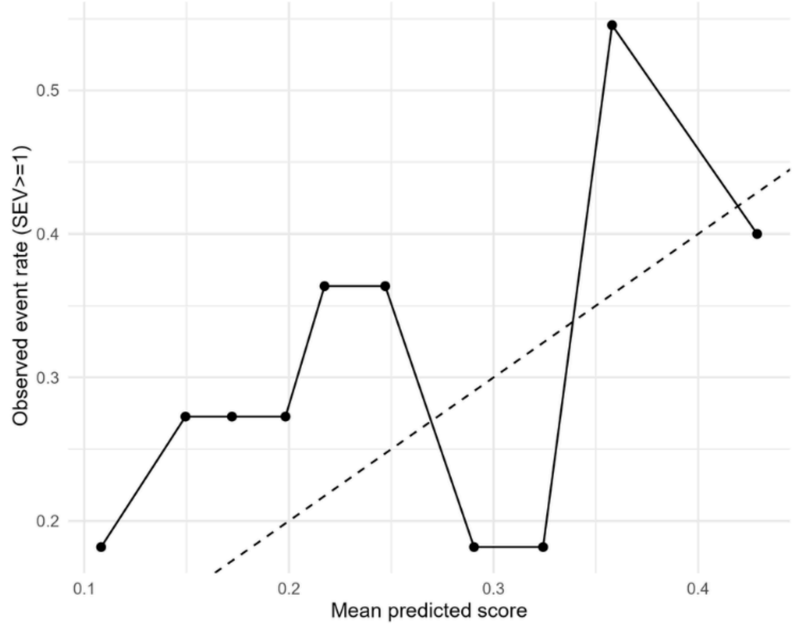
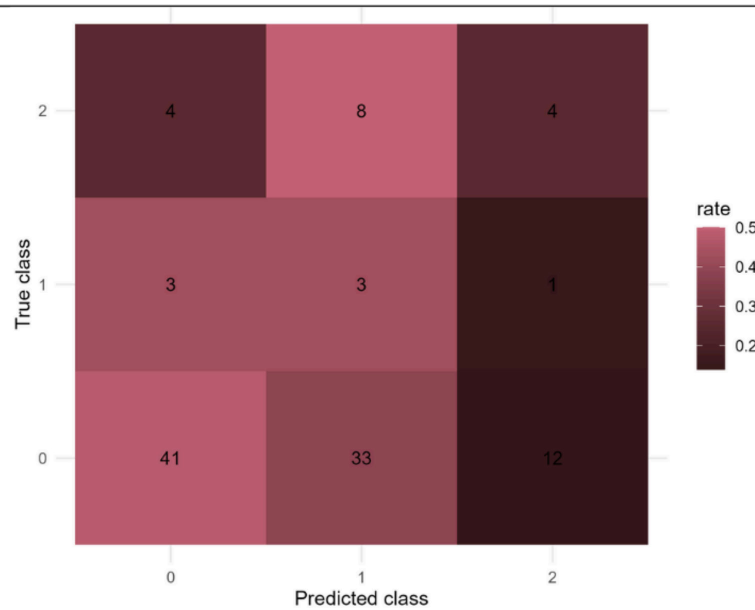



*Figure 1 — Model calibration for severity prediction*
*Reliability diagram showing the relationship between predicted severity risk and observed event frequency (SEV ≥ 1) for the selected model. The dashed line denotes perfect calibration. Deviations at higher predicted risk indicate mild over-confidence in the upper score range.*

*Figure 2 — Confusion matrix for MACE prediction (best model)*
*Confusion matrix of true versus predicted MACE classes for the threshold-optimized ordinal model. Most errors occur between adjacent classes, indicating that misclassifications are primarily near the decision boundaries rather than gross category shifts.*



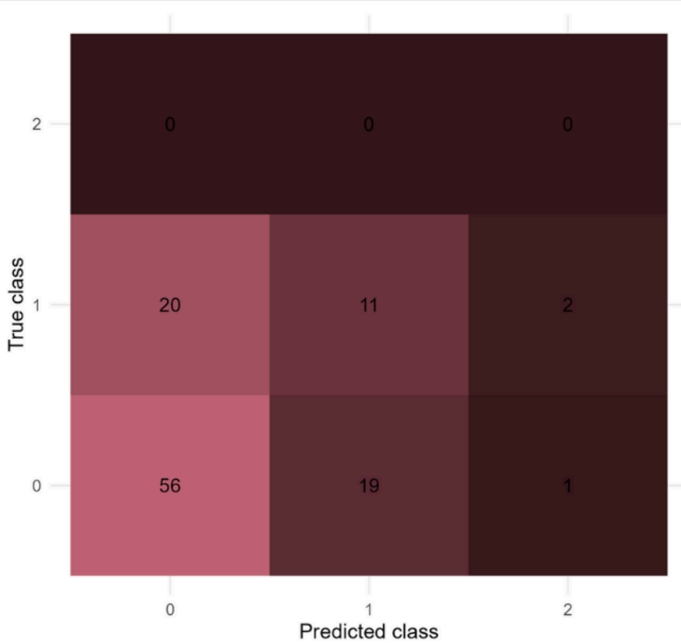




*Figure 3 — Confusion matrix for severity prediction (best model)*
*Confusion matrix of true versus predicted severity classes (0/1/2). The model shows strong separation between no-severity and mild severity cases, with limited confusion into the highest severity class.*

# Appendix

| file | n | mace_qwk | mace_acc | sev_qwk | sev_acc | sev_mode |
|---|---|---|---|---|---|---|
| submission_THRESHOLD_OPT.csv | 109 | 0.1429705 | 0.440367 | 0.1541645 | 0.6146789 | score->3class(prior-matched) |

*Table 1 — Detailed performance metrics of the selected model*
*Summary of sample size, quadratic weighted kappa, accuracy for MACE and severity prediction, and the severity discretization strategy used by the final selected submission.*

| trustii_id | MACE_GT | MACE_PRED | err_mace | SEV_GT | SEV_PRED | SEV_RAW |
|---|---|---|---|---|---|---|
| 31 | 0 | 2 | 2 | 1 | 0 | 0.206510430790075 |
| 24 | 0 | 2 | 2 | 1 | 1 | 0.38744538719325 |
| 138 | 2 | 0 | 2 | 1 | 0 | 0.160192661174537 |
| 19 | 2 | 0 | 2 | 0 | 0 | 0.138262758825726 |
| 36 | 0 | 2 | 2 | 1 | 1 | 0.38432490935398 |
| 15 | 0 | 2 | 2 | 0 | 1 | 0.349817179457871 |
| 133 | 0 | 2 | 2 | 0 | 0 | 0.194368775494174 |
| 72 | 0 | 2 | 2 | 1 | 1 | 0.374032693968576 |
| 114 | 0 | 2 | 2 | 0 | 1 | 0.337638914941366 |
| 48 | 2 | 0 | 2 | 1 | 0 | 0.216620873478281 |
| 92 | 0 | 2 | 2 | 1 | 2 | 0.530796417645801 |
| 4 | 0 | 2 | 2 | 0 | 0 | 0.198674948731068 |
| 74 | 0 | 2 | 2 | 0 | 1 | 0.355248624199343 |
| 29 | 0 | 2 | 2 | 0 | 1 | 0.31254414730771 |
| 136 | 0 | 2 | 2 | 0 | 1 | 0.397121295564558 |

*Table 2 — Highest-impact MACE prediction errors*
*Top failure cases ranked by absolute MACE class error, showing ground-truth labels, predicted labels, and corresponding severity scores. These cases highlight systematic confusion between extreme outcome classes.*

*Table 3 — Highest impact severity prediction errors*
*Top severity misclassification cases ranked by absolute error between predicted and true severity class, including associated MACE labels and continuous severity scores.*

| trustii_id | SEV_GT | SEV_PRED | SEV_RAW | MACE_GT | MACE_PRED | err_sev |
|---|---|---|---|---|---|---|
| 147 | 0 | 2 | 0.453611606126916 | 0 | 1 | 2 |
| 31 | 1 | 0 | 0.206510430790075 | 0 | 2 | 1 |
| 90 | 0 | 1 | 0.347894073343724 | 0 | 1 | 1 |
| 60 | 0 | 1 | 0.311978768229812 | 0 | 0 | 1 |
| 41 | 1 | 0 | 0.110111816498661 | 0 | 0 | 1 |
| 23 | 1 | 0 | 0.136575799530157 | 1 | 2 | 1 |
| 148 | 1 | 0 | 0.240477134653153 | 0 | 0 | 1 |
| 8 | 0 | 1 | 0.438959360434248 | 2 | 1 | 1 |
| 45 | 1 | 0 | 0.213033947557555 | 0 | 0 | 1 |
| 138 | 1 | 0 | 0.160192661174537 | 2 | 0 | 1 |
| 34 | 1 | 0 | 0.278005951142699 | 1 | 0 | 1 |
| 17 | 0 | 1 | 0.395799574961315 | 0 | 1 | 1 |
| 65 | 0 | 1 | 0.346321168066635 | 0 | 1 | 1 |
| 58 | 1 | 0 | 0.209987533525896 | 0 | 1 | 1 |
| 75 | 1 | 0 | 0.128527404897912 | 0 | 1 | 1 |